

BANKHAUL · REQUIREMENTS

BankHaul requirements

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/bankhaul		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	Amish's decisions 14A (far stake driven 3.5 m; R5 status) and 15A (R8 restated), BKH-DDR-003
0.2	2026-10-03	Amish Chadha	TRL 3; targets made measurable, R10 (weak link) and R11 (working distance) added, status from BKH-CAL-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

BankHaul requirements

Measurable requirements for the first prototype station. Each status comes from the calculation note BKH-CAL-001 (docs/04-calcs/01-sizing.md); "on paper" means calculated, not yet tested. R10 and R11 were added at the TRL 2 review (BKH-DDR-001, items 3 and 4). R8 was restated, and R5's status changed, by Amish's decisions of 2026-10-03 (BKH-DDR-003). Amish, 2026-10-03: "i agree with all the 46 recommendations you provided. please proceed."

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 4 or later)	Status at TRL 3
R1	Set and haul a gillnet with nobody entering the water	10 of 10 set and haul cycles completed from the bank	Field trial with partner fishers	Met on paper: once the far stake is set from a boat, no step needs anyone in the water
R2	Working span	At least 30 m (98 ft) between pulleys; stretch goal 50 m (164 ft)	Measured at the trial site	Met on paper: 30 m design case; rope bought for 50 m
R3	Hand pull force	Peak pull under 250 N (56 lbf) to haul a 30 m net with catch	Spring scale or load cell during trials	Met on paper: about 200 N (estimate)
R4	Bank anchor holding	Holds at least 3 kN (675 lbf) in soft wet soil without visible movement	Pull-out test on a CalRig-style rig	Met on paper: screw anchors at factor 2.1 on capacity in soil of 10 kPa shear strength
R5	Offshore stake or float holding	Holds at least 1.5 kN (337 lbf) horizontal load in soft mud	Pull-out tests in a representative bed	Met on paper in soft mud (factor 4.05) and very soft mud (factor 2.03), with the stake driven 3.5 m (decision 14A)
R6	Cycle time	Full haul and reset under 10 minutes for a 30 m net	Timed trials	Met on paper: about 6 minutes (estimate, without clearing fish)
R7	Durability	One fishing season (6 months) of daily use without rope or pulley failure	Season-long pilot with inspection log	Met on paper (estimate): rope factor 5.7 after sun damage
R8	Cost (restated, decision 15A)	Parts under USD 80 per station for a local-materials station, costed from local prices with the co-design partner before the season pilot. The steel station built for the TRL 4 trials is costed and recorded but not held to this figure	Bill of materials for the local-materials station, priced with the co-design partner	Steel trial station: USD 394, accepted for TRL 4. Local-materials station: about USD 98 (first-order estimate), so not yet met; settled by the partner's costing
R9	Level change	Bank anchor relocatable by two people in under 30 minutes as the shoreline moves	Timed relocation	Met on paper: about 22 minutes (estimate)
R10	Weak link	The net's bridle releases between 500 and 700 N, so a net seized by an animal or snag cannot load the loop, stake or fisher beyond that	Pull three samples of each cord batch to release	Met on paper; cord chosen by test when parts are bought
R11	Working distance	The fisher's working spot is at least 5 m (16 ft) from the water's edge at the day's level	Measured at the trial site	Met by layout: 5.0 m

■ Restated by Amish's decisions

- **R8 (decision 15A, 2026-10-03).** Amish: "i agree with all the 46 recommendations you provided. please proceed." The steel station is built for the TRL 4 trials, so they test the method with parts of known strength, and

a local-materials station is costed with the co-design partner. R8 was "Parts under USD 80 per station"; it now applies the USD 80 figure to the local-materials station costed from local prices, and records the steel trial station's cost without holding it to that figure.

- **R5 (decision 14A, 2026-10-03).** The target is unchanged; the far stake is driven 3.5 m, so it is met on paper in very soft mud as well as soft mud.

■ Assumptions

- Fishers can shift from wading-set nets to loop-set nets without losing much catch.
- Most attack hotspots have banks where a screw anchor can hold (soft wet soil of about 10 kPa undrained shear strength or firmer).
- A boat or low-water window exists to place the far stake once per season.
- Nets can be fitted with a bridle at each end to clip to the loop and the bank post.
- Design case: a 30 m by 2 m gillnet, 10 kg of catch and 5 kg of weed; water 1.0 m deep at the far stake.

Safety: R10 and R11 are safety requirements. They limit what a seized net can do to the fisher and keep the fisher back from the water. Neither makes it safe to stand at the water's edge.